

1. Creating and Renaming Files/Directories

Create a directory named `test_dir` using `mkdir`.

Inside `test_dir`, create an empty file called `example.txt`.

Rename `example.txt` to `renamed_example.txt` using `mv`

```
dk@ubuntu:~/Downloads$ ls -ltr
total 0
dk@ubuntu:~/Downloads$
dk@ubuntu:~/Downloads$ mkdir test_dir
dk@ubuntu:~/Downloads$ cd test_dir/
dk@ubuntu:~/Downloads/test_dir$ touch example.txt
dk@ubuntu:~/Downloads/test_dir$ mv example.txt renamed_example.txt
dk@ubuntu:~/Downloads/test_dir$ ls
renamed_example.txt
dk@ubuntu:~/Downloads/test_dir$ pwd
/home/dk/Downloads/test_dir
dk@ubuntu:~/Downloads/test_dir$
```

2. Viewing File Contents

Use `cat` to display the contents of `/etc/passwd`.

Display only the first 5 lines of `/etc/passwd` using `head`.

Display only the last 5 lines of `/etc/passwd` using `tail`.

```

dk@ubuntu:~/Downloads/test_dir$
dk@ubuntu:~/Downloads/test_dir$ cat /etc/passwd | head -5
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
dk@ubuntu:~/Downloads/test_dir$ cat /etc/passwd | tail -5
rtkit:x:117:119:RealtimeKit:/proc:/usr/sbin/nologin
colord:x:118:120:colord colour management daemon:/var/lib/colord:/usr/sbin/nologin
gnome-initial-setup:x:119:65534::/run/gnome-initial-setup:/bin/false
gdm:x:984:984:Gnome Display Manager:/var/lib/gdm3:/bin/false
dk:x:1000:1000:dk:/home/dk:/bin/bash
dk@ubuntu:~/Downloads/test_dir$

```

3. Searching for Patterns

Use grep to find all lines containing the word "root" in /etc/passwd.

```

dk@ubuntu:~/Downloads/test_dir$
dk@ubuntu:~/Downloads/test_dir$ cat /etc/passwd | grep root
root:x:0:0:root:/root:/bin/bash
nm-openvpn:x:107:113:NetworkManager OpenVPN:/var/lib/openvpn/chroot:/usr/sbin/nologin
dk@ubuntu:~/Downloads/test_dir$
dk@ubuntu:~/Downloads/test_dir$

```

4. Zipping and Unzipping

Compress the test_dir directory into a file named test_dir.zip using zip.

Unzip test_dir.zip into a new directory named unzipped_dir.

```

dk@ubuntu:~/Downloads$ ls -ltr
total 4
drwxrwxr-x 2 dk dk 4096 Oct 31 05:42 test_dir
dk@ubuntu:~/Downloads$
dk@ubuntu:~/Downloads$
dk@ubuntu:~/Downloads$ zip -r test_dir.tar test_dir
  adding: test_dir/ (stored 0%)
  adding: test_dir/renamed_example.txt (stored 0%)
dk@ubuntu:~/Downloads$ ls
test_dir  test_dir.tar
dk@ubuntu:~/Downloads$ unzip test_dir.tar -d unzipped_dir
Archive:  test_dir.tar
   creating: unzipped_dir/test_dir/
  extracting: unzipped_dir/test_dir/renamed_example.txt
dk@ubuntu:~/Downloads$ ls
test_dir  test_dir.tar  unzipped_dir
dk@ubuntu:~/Downloads$

```

5. Downloading Files

Use `wget` to download a file from a URL (e.g., <https://example.com/sample.txt>).

```

test_dir  test_dir.tar  unzipped_dir
dk@ubuntu:~/Downloads$ wget https://www.github.com/krdeepu7/DOCTextAI
--2025-10-31 06:52:46-- https://www.github.com/krdeepu7/DOCTextAI
Resolving www.github.com (www.github.com)... 20.207.73.82
Connecting to www.github.com (www.github.com)|20.207.73.82|:443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://github.com/krdeepu7/DOCTextAI [following]
--2025-10-31 06:52:46-- https://github.com/krdeepu7/DOCTextAI
Resolving github.com (github.com)... 20.207.73.82
Connecting to github.com (github.com)|20.207.73.82|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
Saving to: 'DOCTextAI'

DOCTextAI                                     [ <=> ] 220.67K  --.-KB/s  in 0.1s

2025-10-31 06:52:47 (1.48 MB/s) - 'DOCTextAI' saved [225966]

dk@ubuntu:~/Downloads$ ls
DOCTextAI  test_dir  test_dir.tar  unzipped_dir
dk@ubuntu:~/Downloads$

```

6. Changing Permissions

Create a file named `secure.txt` and change its permissions to read-only for everyone using `chmod`.

```
2023-10-31 06:52:47 (1.48 MB/s) = DOCTextAI saved [225900]

dk@ubuntu:~/Downloads$ ls
DOCTextAI test_dir test_dir.tar unzipped_dir
dk@ubuntu:~/Downloads$ touch secure.txt
dk@ubuntu:~/Downloads$ chmod 444 secure.txt
dk@ubuntu:~/Downloads$ ls -ltr
total 236
drwxrwxr-x 2 dk dk 4096 Oct 31 05:42 test_dir
-rw-rw-r-- 1 dk dk 352 Oct 31 06:36 test_dir.tar
drwxrwxr-x 3 dk dk 4096 Oct 31 06:38 unzipped_dir
-rw-rw-r-- 1 dk dk 225966 Oct 31 06:52 DOCTextAI
-r--r--r-- 1 dk dk 0 Oct 31 06:57 secure.txt
dk@ubuntu:~/Downloads$
```

7. Working with Environment Variables

Use `export` to set a new environment variable called `MY_VAR` with the value "Hello, Linux!".

```
-r--r--r-- 1 dk dk 0 Oct 31 06:57 secure.txt
dk@ubuntu:~/Downloads$ export MY_VAR="Hello, Linux!"
dk@ubuntu:~/Downloads$ echo MY_VAR
MY_VAR
dk@ubuntu:~/Downloads$ echo $MY_VAR
Hello, Linux!
dk@ubuntu:~/Downloads$
```

Github link - https://github.com/krdeepu7/DevOps_Learning